

# Yonathan Wiesel

Bnei Brak, Israel | 050-661-8804 | yswiesel@gmail.com | [LinkedIn](#)

## Professional Summary

Full-Stack and High-Performance Software Engineer with a strong foundation in both frontend and backend development, and a specialized focus on performance-critical applications, geospatial data processing, HPC (High Performance Computing), and AI tooling. Proven track record of delivering scalable, secure, and optimized solutions in elite technological environments. Experienced in designing and deploying HPC-ready microservices and systems optimized for real-time geospatial analysis and computational efficiency. Currently completing military service in an elite IDF tech unit; available for new opportunities starting **June 2025**.

## Experience

**Software Developer - IDF Elite Technological Unit 2022 - Present.**

Designed and implemented secure full-stack applications using **React.js**, **Node.js**, **Express.js**, and **Python**.

Built and optimized backend microservices with a focus on high performance, reliability, and security.

Developed asynchronous WebSocket-based systems for real-time geospatial analysis.

Deployed and maintained HPC-friendly systems utilizing **Numba**, **CUDA**, and optimized geospatial algorithms.

Significantly reduced runtime of Line of Sight (LOS) analysis services by over 70% through the use of Numba-accelerated raster processing and inversion of iteration strategy for massive point coverage.

Applied AI tools and frameworks to automate workflows and enhance development efficiency.

Collaborated in fast-paced, mission-critical environments, delivering innovative solutions under tight constraints.

## Technical Skills

**Languages:** JavaScript, TypeScript, Python (Asyncio, Numba, NumPy, Pandas), Java (basic)

**Frontend:** React.js, Redux Toolkit, Vite, Material-UI, HTML, CSS

**Backend:** Node.js, Express.js, RESTful APIs, FastAPI, WebSockets

**Geospatial:** Rasterio, Fiona, ArcGIS, DTM analysis, LOS and RF simulation

**AI Tools:** ChatGPT, Copilot, LangChain, AI-assisted code generation and debugging

**HPC & Optimization:** Numba, CUDA, asynchronous processing, raster masking, matrix optimization

**DevOps & Tools:** Git, Docker, Linux CLI, Zsh, VSCode, Postman

**Soft Skills:** System design, analytical problem-solving, high-pressure decision making

**Languages:** Native Hebrew, Fluent English

## Education

**JBH College - Full-Stack Development Program 2021 - 2022**

Completed intensive hands-on training in full-stack web development and modern frameworks.

**Hebron Yeshiva - Knesset Israel, Jerusalem 2016 - 2019**

Cultivated analytical thinking, disciplined study skills, and rigorous logical reasoning.

## Military Service

**Israel Defense Forces - Elite Technological Unit 2022 - 2025 (Ongoing)**

Full-time software developer in a classified, high-performance technology environment.

Service ends June 2025. Further details and references available upon request.

## Hobbies & Interests

- Deeply engaged in the exploration of advanced AI tools and machine learning frameworks.
- Passion for system architecture, performance engineering, and geospatial intelligence.
- Experienced in building and refining high-throughput computing systems.
- Continually learning new programming paradigms and optimizing development workflows.